

Final exam, ML (MDS + BDMA + MIRI)

11th of June, 2024

- In all your answers, conciseness and clarity will be most appreciated.
- Please justify all your answers.
- All questions have equal weight.
- Exam duration: **1h30m**

Question 1

Choose correct alternative among (is / is not) and complete the following sentences:

1. Feature scaling (is / is not) important in gradient descent because ...
2. Feature scaling (is / is not) important in k-Nearest Neighbors because ...
3. Feature scaling (is / is not) important in the Multilayer Perceptron because ...
4. Feature scaling (is / is not) important in Random Forests because ...
5. Feature scaling (is / is not) important in least squares linear regression because ...
6. Feature scaling (is / is not) important in logistic regression because ...

Question 2

In cross-validation, we use the mean validation error across folds as a metric to optimize hyper-parameters. Apart from computing the mean, we could look into other statistics or aggregates of these validation errors. Can you think of any suitable ones? For each metric you list, please state in what way and why it would be a reasonable one.

Question 3

Whenever possible, invent a binary classification dataset with two numeric input features (you may draw it if you want), such that:

1. Logistic regression achieves perfect classification but a decision tree with maximum depth 1 cannot.
2. Logistic regression achieves perfect classification but a decision tree with maximum depth 2 cannot.
3. Logistic regression cannot separate the two classes, but a decision tree can.
4. Logistic regression separates the two classes, but a decision tree cannot.
5. Neither logistic regression nor a decision tree can separate the classes.